Github link: https://github.com/IanTsai1/CS528-Homeworks

Main files:

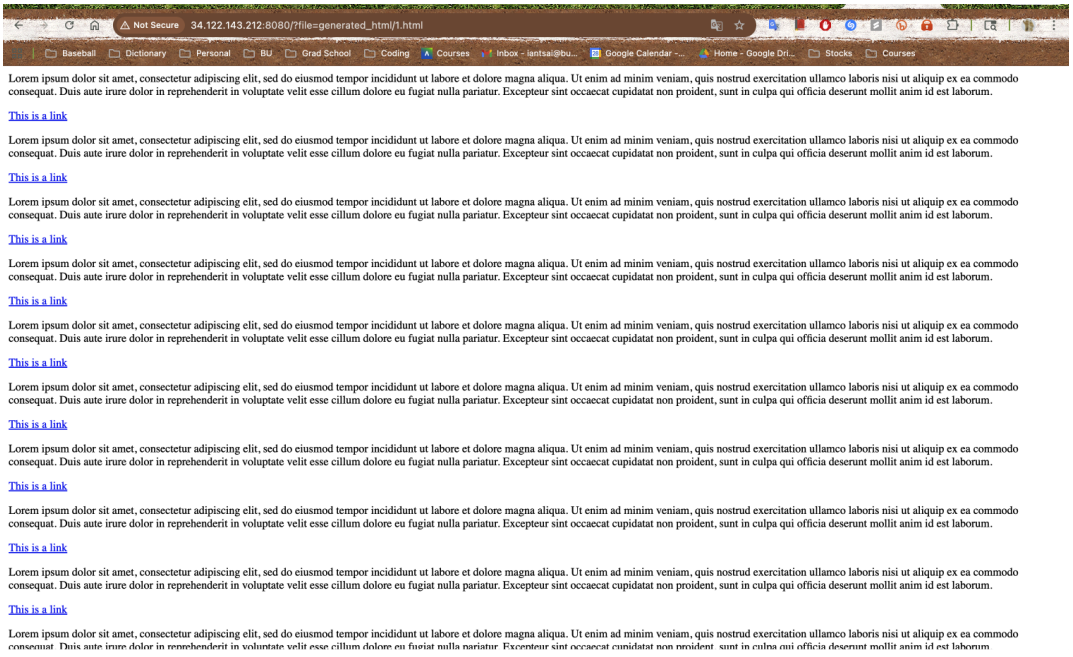
- "hw4/main.py" - web server
- "hw3/report_service.py" - second service code
- "Setup_hw4.sh" - script to start all vm

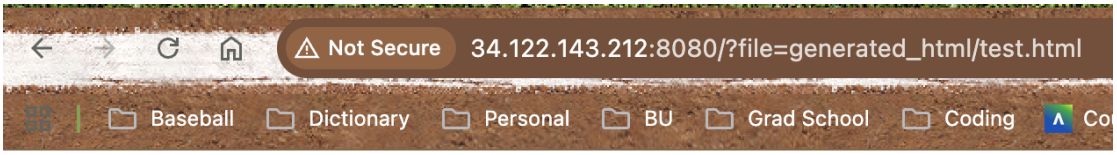
- Steps 1,2,3
  - Setting up virtual machine
    - Run on smallest possible VM
      - Used "e2" machine
    - run this webserver on a VM instance for which you have configured a static IP address
      - Reserved external ip of "34.122.143.212" and set it as external IPv4 address under "Network interfaces"
    - start this web server automatically upon VM creation
      - Wrote a startup script that runs the web server file (main.py) at startup and install necessary dependencies
    - VM runs as a special service account that has only the necessary privileges and nothing more
      - Connect the VM to a newly created service account "webserver-sa@cs528-485121.iam.gserviceaccount.com" that only has logging, pubsub, object viewer, monitoring metric writer permission
      - Monitoring metric writer is needed to get memory related metrics for our VM
  - Code for web server
    - Code in "hw4/main.py"
- Step 4
  - Created a new VM and uploaded linux version of "http-client"
  - Configured the VM so its operating system is Ubuntu 24.04 LTS so that it can run "http-client"
  - './http-client -d 34.122.143.212 -p 8080 -w 'file-service?file=generated_html' -i 20000 -b none -n 100'

-
```
112.46.214.124 - - [01/Mar/2026 05:35:12] "GET / HTTP/1.1" 400 -
34.9.205.64 - - [01/Mar/2026 05:35:12] "GET /file-service?file=generated_html/12217.html HTTP/1.1" 200 -
34.9.205.64 - - [01/Mar/2026 05:35:12] "GET /file-service?file=generated_html/15262.html HTTP/1.1" 200 -
34.9.205.64 - - [01/Mar/2026 05:35:12] "GET /file-service?file=generated_html/16514.html HTTP/1.1" 200 -
34.9.205.64 - - [01/Mar/2026 05:35:12] "GET /file-service?file=generated_html/6931.html HTTP/1.1" 200 -
34.9.205.64 - - [01/Mar/2026 05:35:12] "GET /file-service?file=generated_html/7242.html HTTP/1.1" 200 -
112.46.214.124 - - [01/Mar/2026 05:35:12] "GET / HTTP/1.1" 400 -
34.9.205.64 - - [01/Mar/2026 05:35:12] "GET /file-service?file=generated_html/16485.html HTTP/1.1" 200 -
34.9.205.64 - - [01/Mar/2026 05:35:12] "GET /file-service?file=generated_html/17140.html HTTP/1.1" 200 -
34.9.205.64 - - [01/Mar/2026 05:35:13] "GET /file-service?file=generated_html/17948.html HTTP/1.1" 200 -
34.9.205.64 - - [01/Mar/2026 05:35:13] "GET /file-service?file=generated_html/9945.html HTTP/1.1" 200 -
34.9.205.64 - - [01/Mar/2026 05:35:13] "GET /file-service?file=generated_html/5218.html HTTP/1.1" 200 -
112.46.214.124 - - [01/Mar/2026 05:35:13] "GET / HTTP/1.1" 400 -
34.9.205.64 - - [01/Mar/2026 05:35:13] "GET /file-service?file=generated_html/6953.html HTTP/1.1" 200 -
34.9.205.64 - - [01/Mar/2026 05:35:13] "GET /file-service?file=generated_html/18148.html HTTP/1.1" 200 -
34.9.205.64 - - [01/Mar/2026 05:35:13] "GET /file-service?file=generated_html/3069.html HTTP/1.1" 200 -
34.9.205.64 - - [01/Mar/2026 05:35:13] "GET /file-service?file=generated_html/12803.html HTTP/1.1" 200 -
34.9.205.64 - - [01/Mar/2026 05:35:13] "GET /file-service?file=generated_html/19552.html HTTP/1.1" 200 -
112.46.213.210 - - [01/Mar/2026 05:35:13] "GET / HTTP/1.1" 400 -
34.9.205.64 - - [01/Mar/2026 05:35:13] "GET /file-service?file=generated_html/11659.html HTTP/1.1" 200 -
34.9.205.64 - - [01/Mar/2026 05:35:14] "GET /file-service?file=generated_html/5837.html HTTP/1.1" 200 -
34.9.205.64 - - [01/Mar/2026 05:35:14] "GET /file-service?file=generated_html/3254.html HTTP/1.1" 200 -
34.9.205.64 - - [01/Mar/2026 05:35:14] "GET /file-service?file=generated_html/3505.html HTTP/1.1" 200 -
34.9.205.64 - - [01/Mar/2026 05:35:14] "GET /file-service?file=generated_html/11130.html HTTP/1.1" 200 -
112.46.212.210 - - [01/Mar/2026 05:35:14] "GET / HTTP/1.1" 400 -
34.9.205.64 - - [01/Mar/2026 05:35:14] "GET /file-service?file=generated_html/8927.html HTTP/1.1" 200 -
34.9.205.64 - - [01/Mar/2026 05:35:14] "GET /file-service?file=generated_html/13832.html HTTP/1.1" 200 -
34.9.205.64 - - [01/Mar/2026 05:35:14] "GET /file-service?file=generated_html/15289.html HTTP/1.1" 200 -
34.9.205.64 - - [01/Mar/2026 05:35:14] "GET /file-service?file=generated_html/5900.html HTTP/1.1" 200 -
112.46.214.124 - - [01/Mar/2026 05:35:14] "GET / HTTP/1.1" 400 -
```
  - Output

- Step 5
  - 404
    -
```
~/De/C/CS52/Homeworks  HW2  on  main !3 ?2  curl -i "http://34.122.143.212:8080/?file=generated_html/test.html"
HTTP/1.0 404 Not Found
Server: BaseHTTP/0.6 Python/3.11.2
Date: Sun, 01 Mar 2026 04:53:38 GMT

File not found%
```
    -
```
2026-02-28 23:53:38.394   instance-20260301-0…   File not found in bucket: generated_html/test.html
Investigate log   Copy   Expand nested fields   Hide log summary
{
  insertId: "1try2ijf2dj8ip"
  labels: {1}
  logName: "projects/cs528-485121/logs/python"
  receiveTimestamp: "2026-03-01T04:53:38.454829082Z"
  resource: {2}
  severity: "WARNING"
  sourceLocation: {3}
  textPayload: "File not found in bucket: generated_html/test.html"
  timestamp: "2026-03-01T04:53:38.394671Z"
}
```
  - 501
    -
```
~/De/C/CS52/Homeworks  HW2  on  main !3 ?2  curl -i -X POST  "http://34.122.143.212:8080"
HTTP/1.0 501 Not Implemented
Server: BaseHTTP/0.6 Python/3.11.2
Date: Sun, 01 Mar 2026 04:53:18 GMT

Not implemented%
```

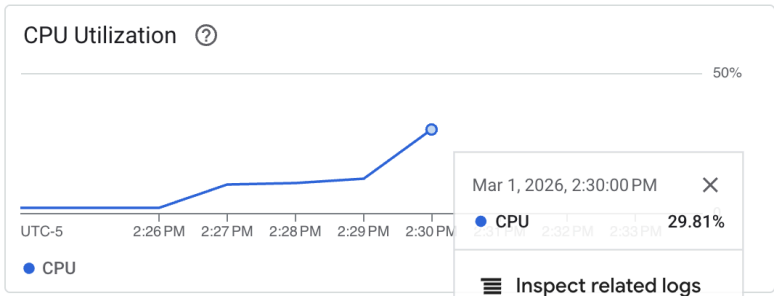
-

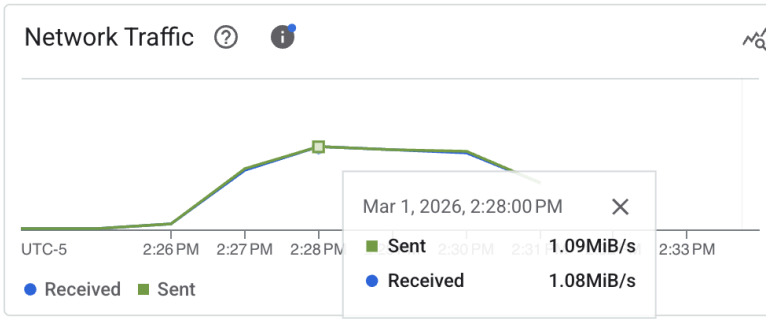
- Step 6
  - 200
    - 

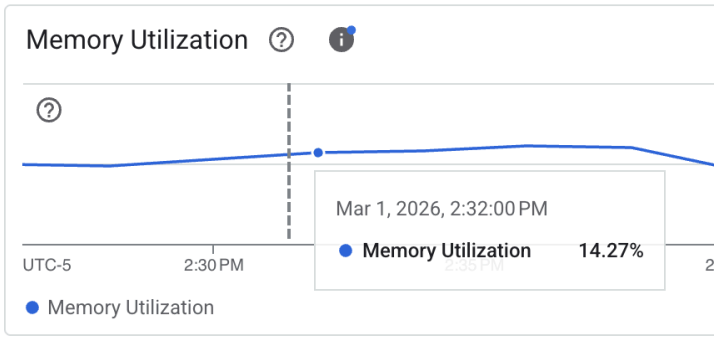
  - 404
    - 

      File not found
- Step 7
  -

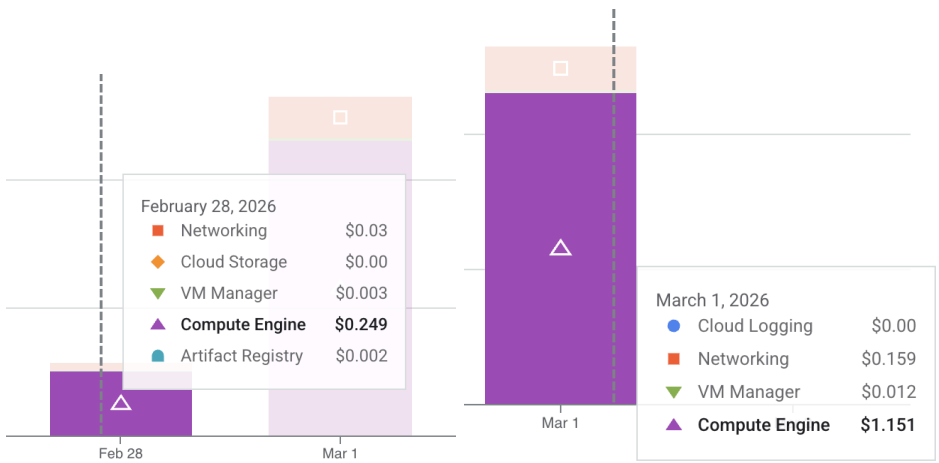
    - Curl with forbidden country header

- ```
  ~/De/C/CS528 - Cloud Computing/Homeworks  HW2  on  main !3 ?2  gcloud compute ssh instance-20260301-160854 --zone=us-central1-c
  Warning: Permanently added 'compute.1604632753025440547' (ED25519) to the list of known hosts.
  Linux instance-20260301-160854 6.1.0-43-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.162-1 (2026-02-08) x86_64

  The programs included with the Debian GNU/Linux system are free software;
  the exact distribution terms for each program are described in the
  individual files in /usr/share/doc/*/copyright.

  Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
  permitted by applicable law.
  Ian@instance-20260301-160854:~$ tail -f /var/log/report_service.log
  Listening for messages on projects/cs528-485121/subscriptions/forbidden-topic-sub...
  REPORT SERVICE ALERT: Forbidden request from Iran
  ```
  - VM2 print output
- ```
  73.61.54.4 - - [01/Mar/2026 16:27:12] "GET /?file=generated_html/1.html HTTP/1.1" 400 -
  ```
  - VM1 logging output
- Step 8
  - Can't saturate using 4 concurrent clients
  - 

  - 

  - 

  - Based on the performance graphs, increasing the number of concurrent clients causes a rise in CPU utilization and network bandwidth as the server processes more simultaneous requests and fetches additional data from

your bucket. The CPU peaked at 29.81% and bandwidth reached approximately 1.09 MiB/s, memory usage remained stable near 14%,
    - There also seems to be some delay in the logging of the metrics, which may be due to batch processing and eventual consistency.
    - As you increase concurrent clients, CPU utilization and network bandwidth scale linearly because the server must dedicate more processing cycles and data throughput to manage the additional simultaneous requests. In contrast, memory usage remains relatively stable because the primary RAM footprint is tied to the fixed overhead of the web server and OS, rather than the volume of active connections.
    - Cost
        - 

        -

Service	Usage cost	Savings programs	Other savings	Subtotal	% Change
Compute Engine	$1.40	–	–	$1.40	New
Networking	$0.19	–	–	$0.19	New
VM Manager	$0.02	–	–	$0.02	New
Artifact Registry	$0.01	–	–	$0.01	-
Cloud Storage	$0.00	–	–	$0.00	-
Cloud Logging	$0.00	–	–	$0.00	-

Subtotal	$1.61
Tax	–
Filtered total	$1.61

            - Usage cost for this hw
- Step 9
    - Make the script executable: 'chmod +x setup_hw4.sh'
    - Run the script './setup_hw4.sh'
    - After running, the web server should be visible in "http://34.122.143.212:8080/?file=generated_html/1.html"
    - I also commented a part of the code that will delete any existing instances since my gcloud has a limit of 4 VM instances at the time
    - Note: Used LLM to assist with script writing